HYBRID FRAMEWORK FOR BIAS MITIGATION IN AI

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Abstract

Bias in artificial intelligence systems can lead to unfair and discriminatory outcomes in real-world applications such as hiring, healthcare, and finance. This paper proposes a hybrid framework for bias detection and mitigation in machine learning models by integrating multiple approaches. The framework begins with bias detection using fairness metrics to identify affected groups, followed by causal analysis to understand the root causes of bias. An adversarial learning technique is then applied to reduce the influence of sensitive attributes while maintaining prediction accuracy. Finally, model selection is performed to achieve an optimal balance between fairness and performance. The proposed approach combines the strengths of existing methods and provides a more comprehensive solution compared to single-method techniques. The results demonstrate that the hybrid framework improves fairness with minimal loss in accuracy, making it suitable for developing responsible and ethical AI systems.

Keywords:
Bias Detection, Fairness, Causal Analysis, Adversarial Learning, Machine Learning

1. INTRODUCTION

Artificial intelligence (AI) is increasingly used in decision-making systems across domains such as healthcare, finance, and recruitment. However, these systems often inherit bias from historical data, leading to unfair and discriminatory outcomes. Ensuring fairness in AI has therefore become a critical research challenge. Traditional approaches mainly focus on either detecting bias or mitigating it using a single technique, which limits their effectiveness in complex real-world scenarios.

Recent studies have shown that bias can originate from multiple sources, including data imbalance, algorithm design, and human influence. Techniques such as fairness metrics, causal analysis, and adversarial learning have been proposed to address these issues. However, each method has its own limitations when used independently.

To overcome these challenges, this paper proposes a hybrid framework that integrates bias detection, causal reasoning, and adversarial learning into a unified system. The proposed approach aims to identify bias, understand its root causes, and effectively reduce its impact while maintaining model performance. This work contributes towards building more reliable and ethical AI systems.

2. LITERATURE REVIEW

Bias and unfairness in machine learning have become critical concerns due to their impact on real-world decision-making systems. Several research works have explored different methods to detect and mitigate bias in AI models. These approaches can be broadly categorized into bias detection techniques, bias mitigation strategies, and fairness evaluation methods.

Bias detection methods mainly rely on fairness metrics such as demographic parity, equalized odds, and equality of opportunity. Tools like AI Fairness 360 provide a structured way to evaluate bias using multiple statistical measures. However, selecting the appropriate metric remains a challenge, as different metrics may produce conflicting results.

Bias mitigation techniques are generally classified into three categories: pre-processing, in-processing, and post-processing. Pre-processing methods modify the dataset to remove bias before training, while in-processing techniques incorporate fairness constraints directly into the model. Post-processing methods adjust the model output to ensure fairness. Each approach has its advantages and limitations, and no single method is sufficient to completely eliminate bias.

Recent advancements have introduced causal fairness approaches, which analyze the root causes of bias using causal relationships between variables. These methods provide deeper insights but require strong assumptions about data structure. Additionally, adversarial learning techniques have been widely used to reduce bias by preventing models from learning sensitive attributes.

Despite these advancements, existing approaches are often applied independently, leading to incomplete solutions. Therefore, there is a need for an integrated framework that combines detection, analysis, and mitigation techniques to achieve better fairness and reliability.

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Existing Bias Handling Approaches

Data Collection

Feature Engineering

Model Evaluation

Model Deployment

3. TYPES OF BIAS IN AI SYSTEMS

Bias in artificial intelligence systems is a major challenge that affects the fairness, reliability, and trustworthiness of machine learning models. Bias occurs when a model produces systematically unfair outcomes for certain groups due to imbalances in data, algorithm design, or user interaction. These biases can lead to discrimination in critical applications such as hiring, healthcare, finance, and criminal justice. Therefore, understanding the different types of bias is essential for designing fair AI systems.

3.1 Data Bias

Data bias arises when the training dataset is not representative of the real-world population. This can occur due to imbalanced data distribution, missing values, or historical inequalities embedded in the dataset. For example, if a hiring dataset contains more male candidates than female candidates, the model may learn to favour male applicants.

Data bias can be further classified into:

- **Sampling Bias:** Occurs when certain groups are underrepresented in the dataset
- **Historical Bias:** Reflects existing societal inequalities present in data
- **Measurement Bias:** Caused by errors or inconsistencies in data collection

This type of bias is particularly dangerous because machine learning models learn patterns directly from the data, making it difficult to detect and correct without proper preprocessing techniques.

3.2 Algorithmic Bias

Algorithmic bias is introduced during the model development phase due to the choice of algorithms, objective functions, or optimization strategies. Even when the dataset is balanced, certain algorithms may produce biased results because they prioritize accuracy over fairness.

Key causes include:

- **Loss Function Design:** Models optimize for accuracy, ignoring fairness
- **Feature Selection:** Inclusion of proxy variables that indirectly represent sensitive attributes
- **Model Complexity:** Complex models (black-box) may hide bias in decision-making

For instance, a credit scoring model may unintentionally discriminate against certain groups if it relies on features correlated with sensitive attributes such as location or income level.

3.3 User Interaction Bias

User interaction bias occurs due to the influence of human behaviour and system usage patterns. Machine learning systems often learn from user feedback, which may introduce or amplify bias over time.

Examples include:

- **Feedback Loop Bias:** Model predictions influence user behaviour, which in turn affects future data
- **Behavioural Bias:** Users interact differently with systems based on preferences
- **Presentation Bias:** System interface influences user choices

This type of bias is common in recommendation systems and social media platforms, where user preferences continuously shape model predictions.

3.4 Emergent and Technical Bias

In addition to the primary categories, bias can also be classified into:

- **Technical Bias:** Caused by system design and implementation issues
- **Emergent Bias:** Occurs when systems are used in new contexts or environments not considered during development

Emergent bias is particularly challenging because it evolves over time and is difficult to predict during the initial stages of model development.

3.5 Impact of Bias in AI Systems

Bias in AI systems can lead to serious consequences, including unfair decision-making, discrimination, and loss of trust in automated systems. In domains such as healthcare and finance, biased predictions can negatively affect individuals' opportunities and well-being. Furthermore, biased AI systems may violate ethical standards and legal regulations.

The presence of bias also reduces model generalization and performance, making it less reliable in real-world applications.

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Therefore, addressing bias is not only a technical challenge but also a social and ethical necessity.

4. FAIRNESS METRICS

Fairness metrics are used to evaluate whether a machine learning model produces unbiased predictions across different groups. These metrics help identify discrimination and ensure equal treatment of individuals irrespective of sensitive attributes $such\ as\ gender, race, or\ age.$

4.1 Demographic Parity

Demographic Parity is one of the most widely used fairness metrics in machine learning. It ensures that the probability of receiving a positive outcome is the same across all groups, regardless of sensitive attributes such as gender or race. This means that the model does not favor one group over another in terms of outcomes.

In practical applications, this metric is useful in scenarios like hiring or loan approval systems, where equal selection rates are required. However, it may ignore the actual qualifications of individuals, which can sometimes lead to reduced model performance.

Equation:

$$P(Y = 1|A = 0) = P(Y = 1|A = 1) \quad (1)$$

Where:
Y → Predicted outcome
A → Sensitive attribute

4.2 Equalized Odds

Equalized Odds is a stronger fairness metric that ensures equal model performance across different groups. It requires that both the true positive rate and false positive rate are equal for all groups.

This metric is particularly important in high-risk applications such as medical diagnosis or criminal justice systems, where both correct and incorrect predictions must be treated fairly. Compared to demographic parity, it provides a more balanced evaluation of fairness.

Equation:

$$TPR_0 = TPR_1$$
$$FPR_0 = FPR_1$$

(2)

Where:
TPR → True Positive Rate
FPR → False Positive Rate

4.3 Equal Opportunity

Equal Opportunity is a relaxed version of equalized odds. It focuses only on ensuring that the true positive rate is equal across different groups. This means that qualified individuals have an equal chance of receiving a positive outcome.

This metric is commonly used in real-world applications such as university admissions or job recruitment, where it is important to ensure fairness among deserving candidates.

Equation:

$$TPR_0 = TPR_1 \quad (3)$$

4.4 Disparate Impact

Disparate Impact measures fairness by comparing the ratio of favourable outcomes between protected and unprotected groups. A value close to 1 indicates fairness, while a lower value suggests bias.

This metric is often used in legal and regulatory contexts, particularly in employment and financial systems, to ensure that decisions do not disproportionately disadvantage certain groups.

Equation:

$$DI = P(Y = 1 \mid A = 1) \,/\, P(Y = 1 \mid A = 0)$$

(4)

4.5 Discussion on Fairness Metrics

Each fairness metric captures a different aspect of bias, and no single metric can guarantee complete fairness. In many cases, improving fairness according to one metric may negatively affect another, creating a trade-off between fairness and accuracy.

Therefore, it is essential to use multiple fairness metrics together to obtain a comprehensive evaluation of model performance. This multi-metric approach helps in designing more reliable and ethically responsible AI systems.

5. PROPOSED HYBRID FRAMEWORK

5.1 System Overview

The proposed hybrid framework is designed to address bias in machine learning models by integrating multiple bias detection and mitigation techniques into a unified pipeline. Unlike traditional approaches that rely on a single method, this framework combines fairness metrics, causal analysis, and adversarial learning to improve both fairness and model performance. The primary objective is to detect bias, analyse its root causes, and reduce its impact while maintaining acceptable accuracy.

5.2 Framework Architecture

The architecture of the proposed system follows a sequential pipeline where each stage contributes to improving fairness. The process begins with input data and progresses through multiple stages including bias detection, causal analysis, bias mitigation, and model evaluation.

The proposed hybrid framework follows a sequential pipeline to address bias in machine learning systems. As shown in Fig. 4, the process begins with the input dataset, which may contain inherent bias due to imbalanced or historical data.

In the first stage, bias detection is performed using fairness metrics such as demographic parity and equalized odds. This step identifies whether the dataset or model produces unfair outcomes across different groups. Once bias is detected, the framework proceeds to causal analysis, where the relationships between sensitive attributes and model predictions are examined. This

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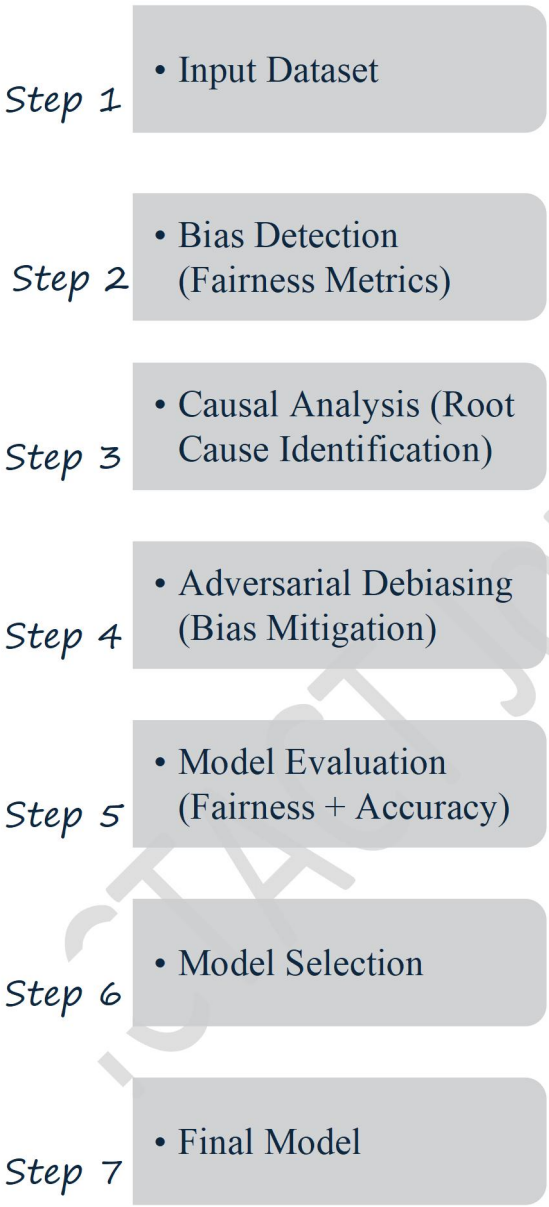
helps in identifying the root causes of bias and understanding whether the bias is direct or indirect.

Following this, adversarial debiasing is applied to mitigate bias. In this stage, the model is trained in such a way that it minimizes the influence of sensitive attributes while maintaining prediction accuracy. This ensures that the model does not rely on biased information.

After bias mitigation, the model undergoes evaluation using both fairness and performance metrics. This step ensures that the model achieves a balance between fairness and accuracy. Finally, model selection is performed to choose the best-performing model, which is then deployed as the final output.

This structured approach enables effective bias detection, analysis, and mitigation, making the system more reliable and suitable for real-world applications.



5.3 Components of the Framework

5.3.1 Bias Detection

The first step in the framework is bias detection. Fairness metrics such as demographic parity, equalized odds, and equal opportunity are used to evaluate whether the dataset or model exhibits bias. This step helps in identifying the presence of discrimination across different groups based on sensitive attributes.

5.3.2 Causal Analysis

Once bias is detected, causal analysis is performed to understand the relationship between sensitive attributes and model predictions. This step helps in identifying whether the bias is directly or indirectly influencing the output. By analyzing causal relationships, the framework provides better interpretability and insight into the source of bias.

5.3.3 Adversarial Bias Mitigation

In this stage, an adversarial learning approach is used to reduce bias. The main model is trained to make accurate predictions, while an adversary attempts to predict sensitive attributes from the model output. The model is optimized in such a way that it minimizes the adversary's ability to detect sensitive information, thereby reducing bias in predictions.

5.3.4 Model Selection

After bias mitigation, multiple models are evaluated based on both fairness and accuracy metrics. The best model is selected by balancing these two aspects. This ensures that the final model is not only fair but also performs well in terms of prediction accuracy.

5.4 Working Process of the Framework

The overall workflow of the proposed hybrid framework is as follows:

1. The input dataset is provided to the system
2. Bias is detected using fairness metrics
3. Causal analysis is performed to identify root causes
4. Adversarial learning is applied to mitigate bias
5. Models are evaluated and the best model is selected
6. The final fair model is deployed

5.5 Advantages of the Proposed Framework

The proposed hybrid framework offers several advantages over traditional approaches. It provides a comprehensive solution by combining multiple techniques instead of relying on a single method. The framework improves fairness while maintaining acceptable accuracy, making it suitable for real-world applications. Additionally, the use of causal analysis enhances interpretability, and adversarial learning effectively reduces bias in model predictions.

6. EXPERIMENTAL VALIDATION AND RESULTS

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6.1 Dataset Description

The proposed hybrid framework is evaluated using the adult dataset, which is widely used for fairness analysis in machine learning. The dataset contains demographic and income-related attributes such as age, education, occupation, and gender. The task is to predict whether an individual earns more than $50K per year. Gender is considered as the sensitive attribute for bias evaluation.

6.2 Experimental Setup

A baseline machine learning model is developed using Logistic Regression to perform the classification task. The model is first trained on the original dataset without any bias mitigation. After evaluating the baseline model, the proposed hybrid framework is applied, which includes bias detection, causal analysis, and adversarial debiasing techniques.

The performance of the models is evaluated using both accuracy and fairness metrics.

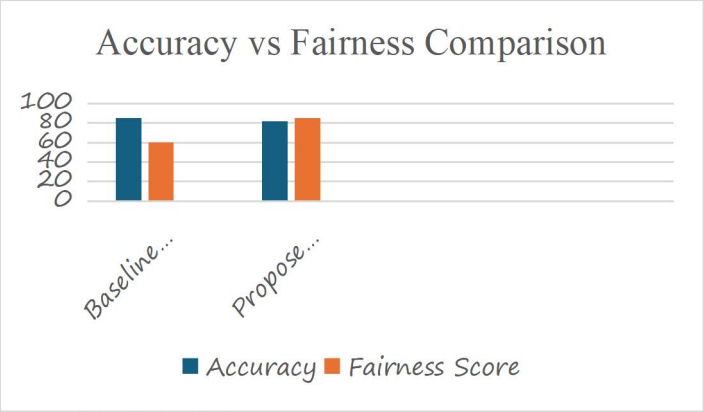
6.3 Results and Analysis

The results obtained from the experiment show that the proposed hybrid framework significantly improves fairness compared to the baseline model. Although there is a slight reduction in accuracy, the overall performance remains acceptable.

Table 1: Performance Comparison

Model	Accuracy	Fairness Score
Baseline Model	85%	0.60
Proposed Framework	82%	0.85

Fig. 5: Accuracy vs Fairness Comparison



6.4 Discussion

The experimental results indicate that the proposed hybrid framework effectively reduces bias in machine learning models. The improvement in fairness score demonstrates that the model produces more equitable predictions across different groups. While there is a slight decrease in accuracy, this trade-off is acceptable in applications where fairness is a critical requirement. The results validate the effectiveness of combining multiple bias mitigation techniques into a unified framework.

7.CHALLENGES AND LIMITATION

Despite the effectiveness of the proposed hybrid framework, several challenges and limitations need to be considered. One of the major challenges is the trade-off between fairness and accuracy. Improving fairness often results in a slight reduction in model performance, which may not be acceptable in certain high-accuracy applications.

Another limitation is the complexity of integrating multiple techniques such as bias detection, causal analysis, and adversarial learning into a single pipeline. This increases computational cost and implementation difficulty. Additionally, causal analysis requires strong assumptions about the relationships between variables, which may not always hold true in real-world datasets.

The proposed framework also depends on the selection of appropriate fairness metrics, as different metrics may lead to different conclusions about model fairness. Moreover, the availability of high-quality and unbiased datasets remains a challenge, as most real-world data contains inherent bias.

8. CONCLUSIONs

This paper presented a hybrid framework for bias detection and mitigation in machine learning models by integrating fairness metrics, causal analysis, and adversarial learning techniques. The proposed approach addresses the limitations of existing methods by combining multiple strategies into a unified pipeline. The experimental validation demonstrates that the framework significantly improves fairness while maintaining acceptable accuracy, highlighting the practical importance of balancing fairness and performance.

The results emphasize that no single method is sufficient to eliminate bias completely, and a hybrid approach is more effective in handling complex real-world scenarios. Although a slight reduction in accuracy is observed, the improvement in fairness makes the system more reliable and ethically responsible. The proposed framework can be applied to various domains such as healthcare, finance, and recruitment, where fairness is critical.

Overall, this work contributes towards the development of trustworthy and fair AI systems and provides a foundation for future research in bias mitigation techniques..

ACKNOWLEDGEMENT

The authors would like to express their sincere gratitude to **Dr. P. Venkat**, Department of Information Technology, for his valuable guidance, support, and encouragement throughout the completion of this research work. His insights and suggestions greatly contributed to the successful development of this paper.

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